NAVRACHANA UNIVERSITY

Teaching Plan

Big Data Analytics and Data Visualization (CSE603)

Credit : 3

Faculty	Name	Contact	Office Hours	Office Location
	Yogesh Chaudhari	yogeshc@nuv.ac.in		

School	SET

Semester	Spring (Jan to June 2026)			
Lecture time & Weedays	SECTION_1	Monday	11:00:00-12:00:00	05-01-2026 TO 08-05-2026
		Wednesday	12:00:00-13:00:00	05-01-2026 TO 08-05-2026
		Friday	15:00:00-16:00:00	05-01-2026 TO 08-05-2026
	SECTION_2	Monday	12:00:00-13:00:00	05-01-2026 TO 08-05-2026
		Tuesday	12:00:00-13:00:00	05-01-2026 TO 08-05-2026
		Friday	14:00:00-15:00:00	05-01-2026 TO 08-05-2026

Pre-requisite	

Co-requisite	

Anti-requisite	

Course Description	This course is aimed at exposing students to the basic concepts of Big Data and associated technologies for the complete Big Data Management Life Cycle namely Big Data Processing, Big Data Storage and Big Data Visualization. Technologies like Apache Hadoop, Map Reduce, HDFS, Apache Spark, MongoDB are discussed.

Course Abstract	This course, "Big Data Analytics and Data Visualization," addresses the critical need to process and interpret massive datasets in the digital age. Beginning with the foundational 5 V's and the architectural challenges of high-volume data, students transition into the technical core of distributed computing. The curriculum explores the industry-standard Apache Hadoop ecosystem for scalable storage and MapReduce processing. It then advances to Apache Spark to master high-speed, in-memory analytics using Resilient Distributed Datasets (RDDs). To handle diverse data types, the course covers NoSQL paradigms and MongoDB implementation, emphasizing the CAP theorem and flexible schema design. Finally, the focus shifts to data visualization, where students apply statistical analysis and design principles to create impactful visual narratives using techniques like heat maps and box plots. By blending data engineering with visual storytelling, this course prepares learners to transform raw, complex information into actionable business insights.

Course Objectives	1. Describe big data and use cases from selected business domains. 2. Explain No Sql big data management. 3. Install, configure, and run Hadoop and HDFS. 4. Perform map-reduce analytics using Hadoop. 5. Use related tools such as Apache Spark and MongoDB for big data analytics. 6. Study various DV chart/diagrams.

Course Outcomes	CO1. Describe Big Data Use Cases in a given business scenario. CO2. Operate NOSQL database management system MongoDB. CO3. Visualize a given big data problem in terms of Map Reduce and implement it on Apache Hadoop Framework. CO4. Implement a given big data solution on Apache Spark Platform. CO5. Work with the HDFS file management system. CO6. Recommend/Suggest an appropriate DV representation/diagram for a given business analytics representation use case.
Course Typology	Theory
Course Outline	The course is structured across five progressive units designed to build technical and analytical proficiency. Unit 1 introduces Big Data fundamentals, including the 5 V's and the limitations of traditional systems. Unit 2 focuses on Apache Hadoop, covering HDFS architecture and the MapReduce programming model. Unit 3 transitions to Apache Spark, emphasizing its in-memory computing advantages and RDD-based operations for real-time analytics. Unit 4 explores the NoSQL landscape, specifically delving into MongoDB's document-oriented architecture, CRUD operations, and the CAP theorem to manage non-relational data. Finally, Unit 5 focuses on Data Visualization, teaching students to perform univariate, bivariate, and multivariate analysis. Students will master a variety of graphical representations—including scatter plots, heat maps, and box and whisker plots—to communicate findings effectively. This structured progression ensures students gain end-to-end expertise, moving from data ingestion and processing to sophisticated visual communication and storytelling.
Pedagogy	Regular lectures, Case Studies & Group Discussion, Simulation & Demonstration, and Experiential / Live Coding
Expectations from Students	1. Code of Conduct & Academic Integrity • Professionalism: Maintain a respectful learning environment. This includes maintaining silence during lectures and ensuring your behavior does not distract or harass others. • Academic Honesty: There is a zero-tolerance policy for plagiarism. All submitted work, assignments, and code must be your own. • Preparation: Arrive at every session having reviewed previous materials and ready to engage with the day's topic. 2. Attendance & Punctuality Attendance is mandatory and will be recorded within the first 10 minutes of every session. • Punctuality Rule: * 0–5 Minutes Late: Recorded as "Present." o 5–10 Minutes Late: Recorded as a "Late Arrival." o After 10 Minutes: Recorded as "Absent." • Late Arrival Penalty: Every three (3) late arrivals will be calculated as one (1) absence. • Eligibility for End-Semester Exam without grade drop | Minimum 80% attendance. Eligibility for Examination | Minimum 50% attendance. Below 50% Attendance - Mandatory re-registration of the course. 3. Absence & Medical Leave • Planned Absences: Notify the instructor via email at least 24 hours in advance. • Unplanned/Emergency Absences: Contact the instructor within 24 hours of the missed class. • Medical Leave: Official proof/documentation must be submitted within three (3) days of your return to class. • Responsibility: It is the student's sole responsibility to catch up on missed content and announcements. 4. Participation & Performance • Active Engagement: Students are expected to actively engage in class discussions and participate in all scheduled reviews. • Assessments: You are expected to turn in all homework/assignments on time and achieve passing marks in both Midterm and Final examinations. • Presentations: Demonstration of your work through in-class presentations is a core requirement of this course. 5. Ethical Use of AI and Public Datasets In the context of Big Data Analytics, the following rules apply to all technical work: • AI as a Collaborator: You may use Generative AI (e.g., Gemini, ChatGPT, Copilot) to debug code, explain concepts, or brainstorm visualization ideas. However, you may not use AI to generate entire reports or complete scripts. • The "Oral Defense" Requirement: You must be able to explain the logic of every line of code you submit. If you cannot explain your code during an in-class review, it will be treated as a violation of academic integrity. • Public Data Attribution: When using datasets from Kaggle, AWS, or other repositories, you must provide a full citation. While the data is public, the analytical insights and processing steps must be your original work. • Disclosure Statement: All major project submissions must include an "AI Disclosure Statement" detailing which tools were used and for what specific tasks. • Anti-Collusion: While high-level discussion with peers is encouraged, sharing code files or NoSQL schemas is strictly forbidden and considered collusion. 6. Generative AI Disclosure Statement For all major project submissions, you must include an "AI Disclosure Statement" at the end of your report, detailing: 1. Which AI tools were used (if any). 2. Which specific tasks the AI assisted with (e.g., "AI was used to debug the Spark-MongoDB connector"). 3. A certification that the final analysis and conclusions are the student's original work. +------------------+--+ | Activity | AI/Resource Policy | +------------------+--+ | Weekly Labs | Permitted for debugging and syntax help. | +------------------+--+ | Quizzes/Exams | Strictly Prohibited. | +------------------+--+ | Final Project | Permitted for logic brainstorming; | | | must be disclosed. | +------------------+--+ | Data Visualization | Must cite all external data | | | sources and design inspirations. | +------------------+--
Assessment / Evaluation	
Project / Assignment Details	Students will complete a comparative lab measuring Spark's scalability against standard Python, followed by a NoSQL task focused on optimizing MongoDB schemas for real-time data. They will also perform exploratory analysis on large datasets to identify hidden patterns. The course concludes with a group project involving the ingestion of 5GB+ of data into HDFS, processing with Spark MLlib, and presenting findings via an interactive dashboard. The AWS Academcy course(s) is(are) recommended to gain in-depth insights.
Course Material	Text books: 1. Practical Big Data Analytics, Nataraj Dasgupta, Packt Publishing 2. Big Data Analytics, Venkat Ankam, Packt Publishing 3. Fundamentals of Data Visualization, Claus Wilke, Oreilly Publishing Reference books: 1. Big Data Analytics for Market Intelligence, Darwish & Dina, Packt Publishing 2. Effective Cyber Security: William Stallings, Addison Wesley Publishers
Additional Information	• Software & Hardware Requirements: Students are expected to have a laptop with at least 8GB RAM and 50GB of free disk space to run local clusters. We will primarily use open-source environments like Ubuntu/Linux, Docker, and Databricks Community Edition. • Dataset Resources: For projects, students are encouraged to explore repositories such as Kaggle, UCI Machine Learning Repository, and AWS Open Data Registry. • The "Visualization Portfolio": Beyond passing grades, students are encouraged to document their final dashboard projects on platforms like GitHub or Tableau Public to build a professional portfolio for future job placements. • Design Ethics: Students must adhere to the "Principle of Truthful Data." Any manipulation of visual scales to exaggerate findings will be penalized as a breach of professional ethics.

Session Plan	Topic Title	Session No.	Topic & Subtopic Details	Readings, Cases, etc.	Activities	Important Dates
	Unit 1: Introduction	1	Concept of Big Data: Origins and Evolution	Dasgupta (Ch 1)	Brainstorming: "What data did you generate today?"	
		2		Ankam (Ch 1)	Case Study: Analyzing the 5 V's of Uber/Netflix.	
		3	Types of Big Data: Structured vs. Unstructured	Dasgupta (Ch 1)	Identifying data formats in a sample dataset.	
		4	Semi-structured Data: XML and JSON formats	Ankam (Ch 1)	Practice: Parsing a simple JSON file.	
		5	Popular Applications of Big Data Analytics	Ankam (Ch 2)	Group discussion on Big Data in Healthcare.	
		6	Issues and Challenges in Big Data Solutions	Dasgupta (Ch 1)	Debate: Privacy vs. Personalization.	
	Unit 2: Hadoop & MapReduce	7	Concept and Need for Apache Hadoop Framework	Ankam (Ch 3)	Intro to Distributed Systems vs Centralized.	
		8	Hadoop Eco System: Component Overview	Ankam (Ch 3)	Mapping tools to stages of data pipeline.	
		9	Hadoop Basic Architecture: NameNode/DataNode	Dasgupta (Ch 2)	Diagram labeling.	
		10	Installation of Hadoop: Prerequisites	Dasgupta (Ch 2)	Step-by-step installation guide review.	
		11	Operational Modes: Standalone & Pseudo-distributed	Dasgupta (Ch 2)	Configuration file walkthrough (xml files).	
		12	HDFS Architecture: Blocks and Replication	Ankam (Ch 4)	Calculation: Storage required for replicated data.	
		13	HDFS Operations and CLI Commands	Ankam (Ch 4)	Hands-on: File upload/download via terminal.	
		14	MapReduce Concept: The Programming Model	Dasgupta (Ch 3)	Logic building: Understanding Key-Value pairs.	
		15	MapReduce Example: WordCount walkthrough	Dasgupta (Ch 3)	Manual trace of a WordCount execution.	
		16	Constraints of Hadoop/HDFS/MapReduce	Ankam (Ch 4)	Discussion: When is Hadoop not the answer?	
	Unit 3: Apache Spark	17	Concept and Need for Apache Spark	Dasgupta (Ch 4)	Comparison: Spark vs. MapReduce.	
		18	Spark Architecture: Drivers and Executors	Dasgupta (Ch 4)	Architecture mapping.	
		19	Spark Installation and Environment Setup	Ankam (Ch 5)	Verifying Spark Shell (PySpark/Scala).	
		20	Operational Modes: Local vs. Cluster Mode	Ankam (Ch 5)	Resource allocation discussion.	
		21	Spark Basic Operations: Reading Data	Dasgupta (Ch 4)	Loading CSV/Text files into Spark.	
		22	Introduction to RDD (Resilient Distributed Datasets)	Ankam (Ch 5)	Understanding Immutability and Partitioning.	
		23	RDD Programming: Transformations (Map, Filter)	Ankam (Ch 5)	Coding: Applying transformations to a list.	
		24	RDD Programming: Actions (Collect, Reduce, Count)	Ankam (Ch 5)	Coding: Triggering Spark jobs with actions.	
		25	Directed Acyclic Graph (DAG) in Spark	Dasgupta (Ch 4)	Visualizing the execution plan of a Spark job.	
		26	Lazy Evaluation Concept in Spark	Dasgupta (Ch 4)	Experiment: Observing when code actually runs.	
		27	Key-Value Pair RDDs and Joins	Ankam (Ch 5)	Practice: Joining two datasets in Spark.	
		28	Spark Persistence and Caching Strategies	Dasgupta (Ch 4)	Optimization: Comparing runtimes with cache.	
	Unit 4: NoSQL & MongoDB	29	Concept and Need for NoSQL; Application Areas	Ankam (Ch 6)	Market analysis: Why NoSQL is booming.	
		30	NoSQL Architectures: Key-Value, Document, Column	Ankam (Ch 6)	Identifying use-cases for each type.	
		31	Differences: Conventional RDBMS vs. NoSQL	Dasgupta (Ch 5)	Comparison Table creation.	
		32	ACID vs. BASE Systems	Ankam (Ch 6)	Logic session: Availability vs Consistency.	
		33	CAP Theorem and its detailed working	Ankam (Ch 6)	Case studies of CP, AP, and CA systems.	
		34	MongoDB Architecture and Basic Components	Dasgupta (Ch 5)	Understanding BSON vs JSON.	
		35	MongoDB Operations: Collection & Document Creation	Dasgupta (Ch 5)	Lab: Creating your first database and collection.	
		36	MongoDB Operators: Querying and Updating	Dasgupta (Ch 5)	Practice: Using $set, $inc, and $push.	
		37	MongoDB Advanced Features: Indexing	Ankam (Ch 6)	Performance Test: Querying with/without Index.	
		38	Aggregation Framework in MongoDB	Dasgupta (Ch 5)	Building a data pipeline with $match and $group.	
	Unit 5: Visualization	39	Concept and Need for Data Visualization	Wilke (Ch 1)	Presentation: Impact of visualization on business.	
		40	Effective Data Visualization Tips and Techniques	Wilke (Ch 2)	Critique: Analyzing "bad" charts from the web.	
		41	Univariate Analysis: Bar and Line Charts	Wilke (Ch 3)	Practical: Plotting frequency distributions.	
		42	Bivariate Analysis: Scatter Plots and Correlations	Wilke (Ch 4)	Practical: Identifying trends in 2D space.	
		43	Multivariate Analysis: Heat Maps and Bubble Charts	Wilke (Ch 5)	Practical: Visualizing complex relationships.	
		44	Box and Whisker Plots for Statistical Analysis	Wilke (Ch 6)	Interpreting quartiles/outliers.	
		45	Guidelines for Use Cases & Representation selection	Wilke (Ch 9)	Project: Selecting the right chart for a final dataset.	
Publish Status	Published					